

Hector Hernandez

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Summary

ML/Agentic AI Engineer with 10 years in Intel production environments and hands-on experience building LLM-based tools, multi-agent systems, and scalable data platforms. I design reliable agentic systems that combine models, tools, and data into end-to-end multi-step workflows with explicit planning, structured outputs, tool use, failure handling, and monitoring and observability. Currently completing an MS in Computer Science at ASU while applying LLM-assisted coding, evaluation, and latency/cost/accuracy tradeoffs to ship practical AI systems.

Education

Arizona State University

Expected December 2026

MS in Computer Science (GPA: 3.88/4.0)

- **Coursework:** Agentic AI, Statistical Machine Learning, Artificial Intelligence, Data Processing at Scale, Advanced Operating Systems, Data Structures and Algorithms, Applied Cryptography, Data Visualization.

Arizona State University

Aug 2012 – May 2015

BS in Chemical Engineering

Experience

Data Engineering Intern

Atlanta, GA

Honeywell

May 2026 – August 2026

- Building Python and SQL data pipelines on Databricks across Azure and GCP data lakes that support analytics and reporting workflows across business units, with emphasis on scalable data models and downstream reliability.
- Deploying pipelines and services through CI/CD on Kubernetes, automating build, test, and release to improve reliability and reduce manual intervention.
- Developing LLM-based agents for industrial engineering data analysis, combining tool use and structured outputs to automate exploration and surface insights from large operational datasets.
- Collaborating with engineering teams to improve data quality, automate manual processing, and ship maintainable backend systems.

Software Application Engineer

Chandler, AZ

Intel

June 2020 – July 2025

- Designed end-to-end architecture for Intel's global EDI manufacturing data platform, integrating Python prototypes, Angular/D3/Highcharts visualizations, and backend services for thousands of engineering users.
- Re-architected a legacy Python/Oracle ETL stack into C#/SQL Server, making cost/performance tradeoffs that improved scalability, maintainability, and long-term operational efficiency.
- Optimized database structures and query paths across services, reducing data transfer latency from 30+ seconds to under 3 seconds for production engineering workflows.
- Built Airflow, Docker, Python, and SQL Server pipelines with clear failure handling and reduced manual intervention, improving reliability of recurring manufacturing data workloads.
- Maintained globally distributed production infrastructure, resolving bugs and enhancements while strengthening monitoring, observability, and availability for manufacturing teams.
- Led code reviews and Python standards for the EDI development team, using LLM-assisted coding and structured review habits to improve quality, testability, and defect prevention.

Photolithography Process Engineer

Chandler, AZ

Intel

June 2015 – May 2020

- Sustained and improved Photolithography processes to meet manufacturing demand for throughput, cycle time, and yield.
- Improved quality levels by reducing scrap, minimizing yield loss mechanisms, and enhancing process control.
- Identified and executed cost-reduction solutions aligned with factory budget goals.
- Collaborated with Manufacturing, Maintenance, and Process teams to optimize the overall photolithography area, working extensively with both Nikon and ASML scanners.
- Reduced defectivity mechanisms through targeted engineering solutions and tighter process control.
- Developed and deployed automation tools, including 1-click reports, SQL queries, JSL scripts, and Python scripts to

extract, process, and visualize large datasets from legacy and modern systems.

- Trained and mentored multiple new engineers, providing hands-on guidance to help them understand semiconductor equipment and processes, accelerating their ramp-up in the photolithography area.

AI Projects & Research

VoiceValet

[GitHub](#) 

- Built an agentic phone agent using Claude, Vapi, Deepgram, and Resend to execute outbound calls, integrate tools, handle multi-step tasks, recover from failures, and return transcript summaries by email.

ConvergeAI

[GitHub](#) 

- Built a multi-model consensus solver that runs GPT and Claude in parallel, compares disagreement, refines answers, caches responses, and tracks cost for latency/cost/accuracy tradeoffs.

Claude MCP Server Collection

[GitHub](#) 

- Created 5 MCP servers for stocks, weather, news, Gmail, and PostgreSQL finance data, extending Claude with real-time API access, read-only queries, OAuth actions, and graceful error handling.

Nexus Research

[GitHub](#) 

- Built a multi-agent research pipeline that decomposes questions, retrieves public API context, checks reliability, and emits structured Markdown reports.

τ -bench Research

CSE 598 Research

- Evaluated agentic tool use across Qwen3 and Gemma4 families using τ -bench tasks, error taxonomies, $pass^k$ metrics, vLLM, SLURM, and HPC.

Technologies

Agentic AI & LLMs: Claude API, OpenAI API, MCP, LangChain, Ollama, vLLM, ReAct, Function Calling, multi-agent orchestration, tool-use evaluation

Backend & Systems: Python, FastAPI, Docker, Kubernetes, PostgreSQL, SQL Server, Airflow, Databricks, CI/CD, C#, .NET

ML & Evaluation: scikit-learn, Pandas, NumPy, τ -bench, WildToolBench, $pass^k$ metrics

Dev & Infra: Git, GitHub, Claude Code CLI, SLURM, HPC (A100 / AMD ROCm), VS Code

Supporting: JavaScript, TypeScript, React, Angular, D3.js